

Snowstorms, Visibility, and Public Complaints: Modeling Dog Waste Complaints in New York City Following Major Snow Events

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Abstract

Snowstorms significantly disrupt urban systems, alter pedestrian movement, and influence interactions between residents and municipal service systems. This study investigates how major snowstorms affect dog waste complaints submitted through the NYC 311 complaint system. Using publicly available NYC Open Data records, daily dog waste complaint counts were analyzed across three major snowstorm periods: January 2022, January 2026, and February 2026. Poisson regression models were initially estimated to evaluate whether complaint patterns significantly differed before, during, and after storm events. Diagnostic testing revealed substantial overdispersion across models, requiring quasi-Poisson adjustments. The findings demonstrate that most snowstorms were associated with increases in post-storm complaints, supporting the theory that snowfall temporarily conceals environmental waste before exposure increases during melting periods. However, the February 2026 storm produced a significant decrease in post-storm complaints, indicating that complaint behavior varies across events. The study demonstrates the usefulness of 311 complaint systems for analyzing environmental disruptions and urban behavioral patterns.

Introduction

Snowstorms create complex disruptions within urban environments. In cities such as New York City, snowfall affects transportation systems, pedestrian movement, sanitation operations, and public behavior simultaneously. These disruptions extend beyond the duration of active snowfall and continue throughout post-storm recovery periods.

One understudied aspect of winter storms is their effect on sanitation-related complaints. Dog waste complaints provide a unique lens through which to study environmental visibility and reporting behavior because snowfall may temporarily conceal waste accumulation before exposing it during melting periods.

This study investigates how dog waste complaints changed before, during, and after major snowstorms in New York City. Using NYC 311 complaint data, the project evaluates whether storm periods significantly predict complaint volume changes.

The project contributes to broader discussions about environmental disruption, citizen reporting behavior, and urban governance by examining how public complaints respond to changing environmental conditions.

Literature Review

Research on 311 complaint systems demonstrates that public service requests provide insight into citizen-government interaction and neighborhood conditions. Minkoff (2016) argues that NYC 311 data reflect both municipal performance and public expectations regarding city services.

Additional scholarship suggests that complaint systems are influenced by environmental visibility and accessibility. Hsu, Lee, and Wang (2019) found that citizen reporting behavior changes during periods of environmental disruption and uncertainty.

Snow management has historically been viewed as both a logistical and political issue in New York City. Savas (1973) described snow removal as a highly visible test of municipal effectiveness because disruptions affect public safety, mobility, and sanitation simultaneously. More recent work by Call (2020) reframes winter storms as “snow events,” emphasizing their prolonged and cascading effects on urban systems rather than viewing them as isolated weather incidents.

This study builds upon existing scholarship by examining sanitation complaints connected specifically to dog waste visibility following snowfall events.

Research Question

Does snowfall significantly increase the number of dog droppings complaints in NYC in the 30 days following a major snowstorm?

Hypothesis

The study hypothesizes that complaint patterns changed significantly after snowfall events and that the February 2026 storm may reflect improved post-storm conditions.

Null Hypothesis

The null hypothesis states that there is no significant difference in the mean number of daily dog waste complaints before and after major snowstorms.

Data and Methods

This study uses publicly available NYC Open Data 311 Service Request records. The dataset contains detailed information about non-emergency complaints submitted by residents, including complaint type, descriptors, dates, agency information, and geographic variables. The dataset was filtered to isolate dog waste related complaints. Complaint creation dates were converted into standardized date variables using the lubridate package in R.

Three storm events were analyzed and selected by reviewing NOAA data and those were the January 2022, January 2026 and the February 2026 storm. Each storm period was divided into: Pre-Storm (30 days before snowfall), During Storm and Post-Storm (30 days after snowfall). Daily complaint counts were aggregated for statistical analysis. Because the dependent variable represents count data, Poisson regression models were initially estimated using the following specification:

count ~ storm_period

The Pre-Storm category served as the reference group. However, Poisson regression assumes that the mean and variance of the dependent variable are equal. Diagnostic testing revealed substantial overdispersion across all models.

For the January 2022 storm:

- Mean complaint count = 7.20
- Variance = 20.92
- Dispersion statistic = 2.51

For the January 2026 storm:

- Mean = 26.93
- Variance = 959.89
- Dispersion statistic = 18.58

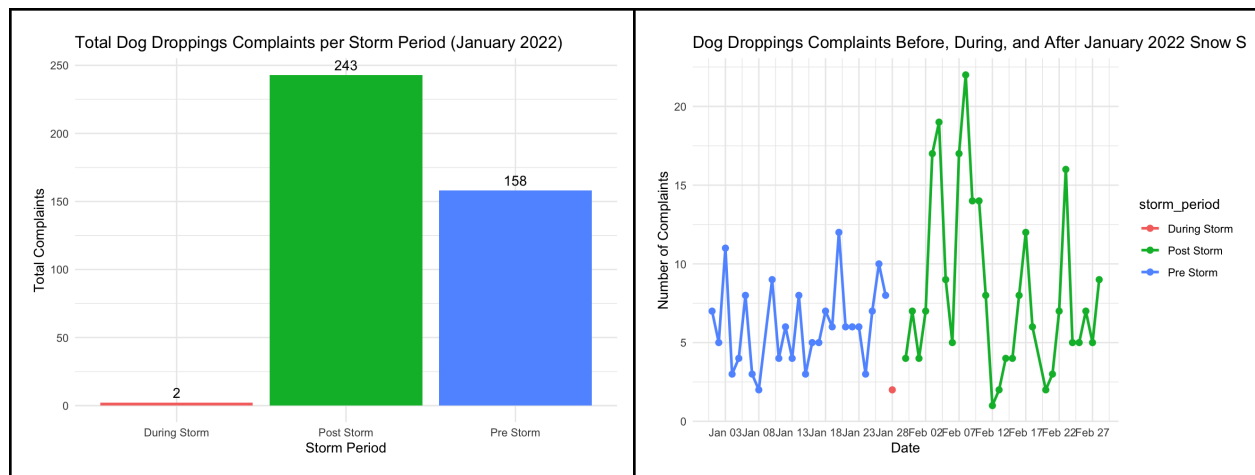
For the February 2026 storm:

- Mean = 27.63
- Variance = 826.88
- Dispersion statistic = 20.42

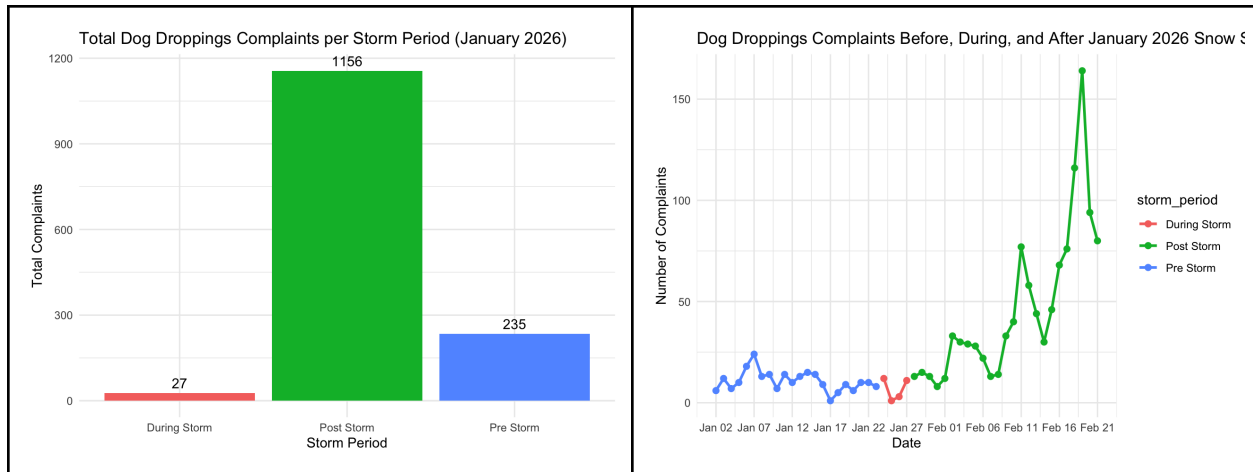
Since the variance substantially exceeded the mean in all models, quasi-Poisson models were estimated to correct for overdispersion and produce more reliable standard errors.

Exploratory Data Analysis

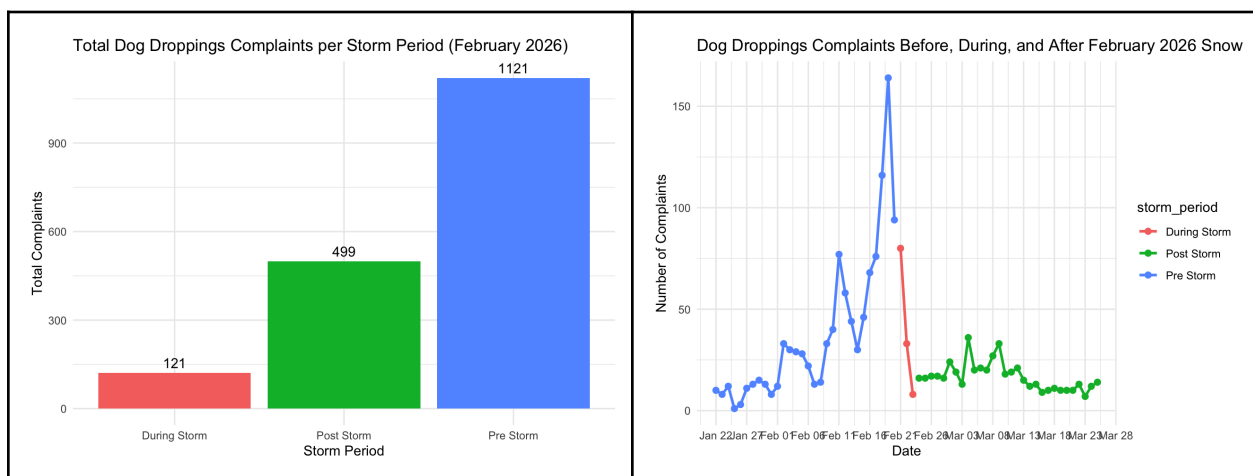
Exploratory data analysis revealed substantial differences across storm events.



The January 2022 and January 2026 storms both displayed substantial increases in complaints after snowfall. Complaint activity remained relatively low during active snowfall periods before increasing during post-storm recovery periods.



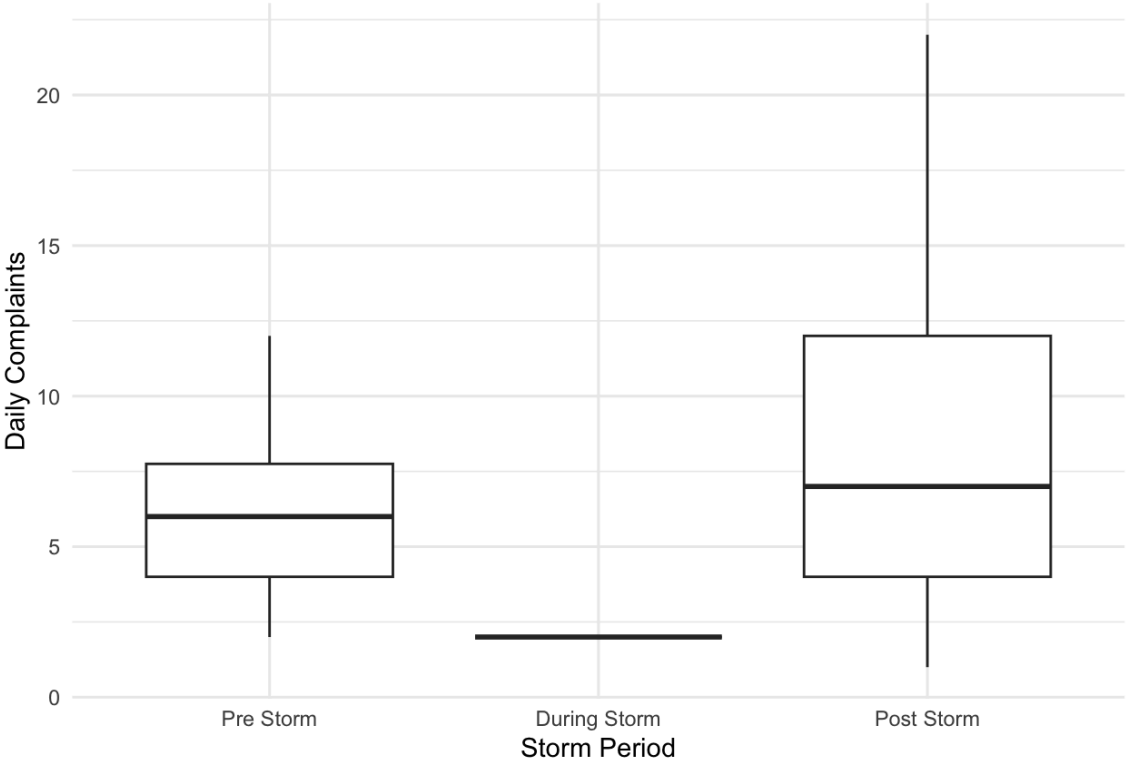
The January 2026 storm produced the most dramatic increase in complaints, suggesting that accumulated snow may have concealed dog waste before exposure increased after melting.



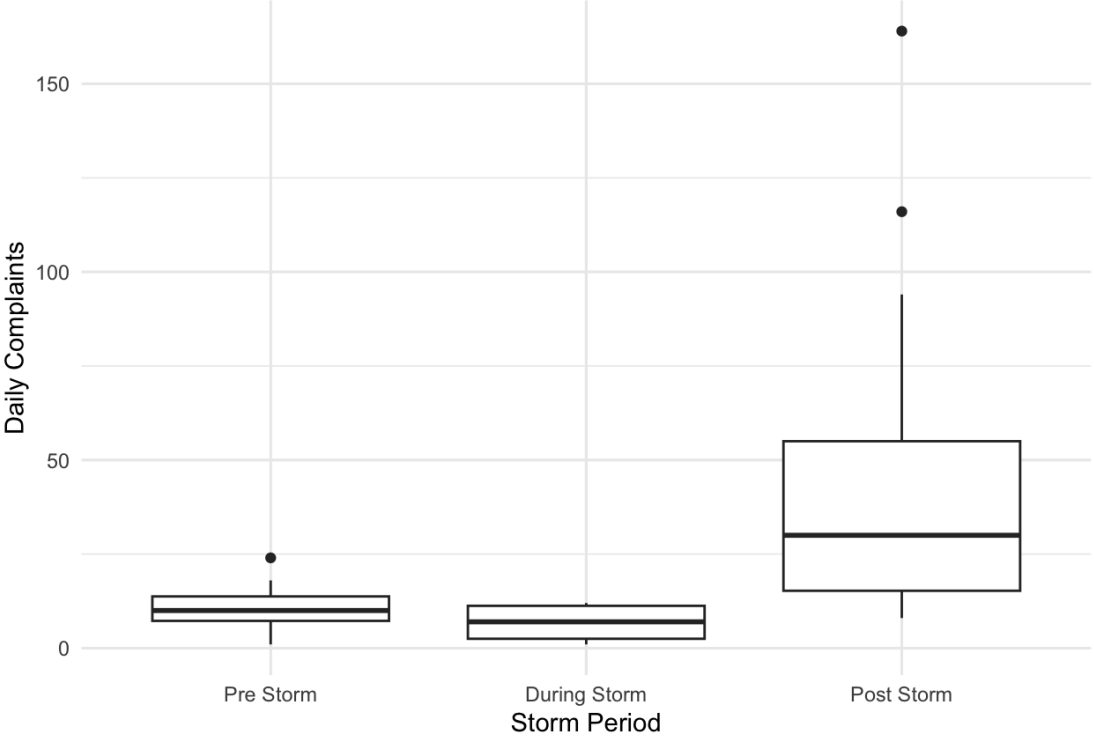
In contrast, the February 2026 storm produced lower complaint totals after snowfall, indicating a substantially different reporting pattern.

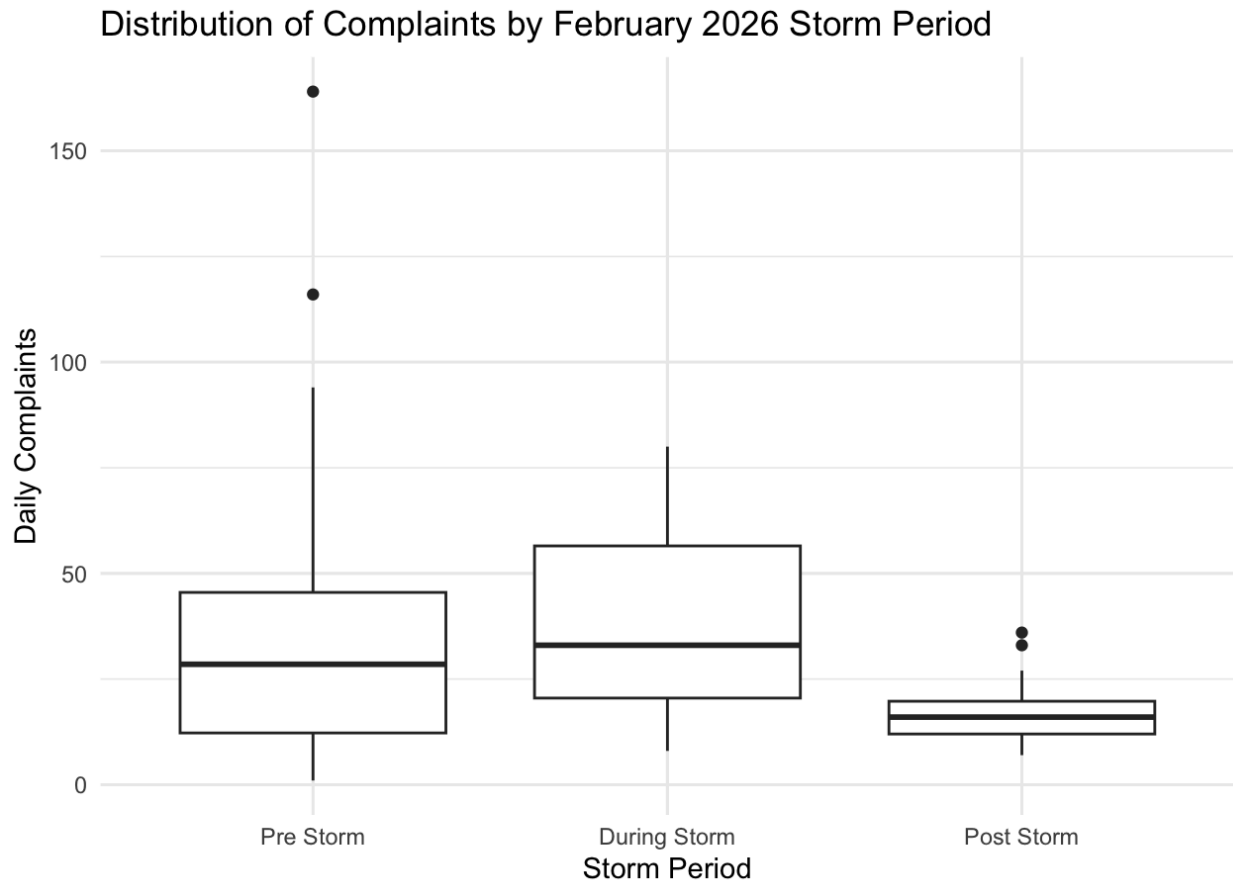
Boxplot visualizations reinforced these findings by showing increased variability and extreme outliers during the January 2026 post- storm period.

Distribution of Complaints by January 2022 Storm Period



Distribution of Complaints by January 2026 Storm Period





Results

The January 2022 Poisson regression model produced statistically significant post-storm effects. The Post-Storm coefficient was: $\beta = 0.321$ and $p = 0.00167$. The Incident Rate Ratio was 1.38, indicating that post-storm complaints increased by approximately 38%. The During-Storm coefficient was negative but not statistically significant: $\beta = -1.111$ and $p = 0.118$. The model fit statistics included: Null deviance = 143.49, Residual deviance = 128.08 and AIC = 338.94. The analysis of deviance tests showed that storm period significantly improved model fit: $p = 0.0004513$. However, overdispersion diagnostics indicated that the Poisson model assumptions were violated. The quasi-Poisson model produced corrected standard errors and showed: Post-Storm coefficient = 0.321 and $p = 0.0524$. Although the significance level weakened slightly after correction, the overall pattern remained consistent.

The January 2026 storm produced much stronger effects. The quasi-Poisson model showed: During-Storm coefficient = -0.459 , $p = 0.602$ and IRR = 0.63. The Post-Storm coefficient was: $\beta = 1.358$, $p < 0.001$ and IRR = 3.89. This indicates that post-storm complaints increased by approximately 289%. The model fit statistics showed: Null deviance = 1406.24, Residual deviance = 835.73 and AIC = 1102.89. The analysis of deviance tests confirmed that storm period significantly improved model fit: $p < 2.2e-16$.

The February 2026 storm produced substantially different results. The quasi-Poisson model showed: During-Storm coefficient = 0.076, $p = 0.860$ and IRR = 1.08. The Post-Storm coefficient was: $\beta = -0.809$, $p = 0.0015$ and IRR = 0.45. This indicates that complaint counts decreased by approximately 55% following the storm. Model fit statistics showed: Null deviance = 1303.89, Residual deviance = 1042.64, and AIC = 1351.58. The storm period variable again significantly improved model fit.

Discussion

The findings suggest that snowstorms significantly influence dog waste complaint behavior in New York City. The January 2022 and January 2026 storms both produced increases in complaints after snowfall. These results support the theory that snow temporarily conceals environmental waste before exposure increases during melting periods. The January 2026 storm produced an especially large increase in complaints, with post-storm counts rising by approximately 289%. This suggests that severe snow accumulation may intensify delayed reporting behavior.

The February 2026 storm produced a substantially different pattern. Rather than increasing after snowfall, complaints declined significantly during the post-storm period. Several factors may explain this difference: differences in storm severity, faster snow clearing, changes in pedestrian movement, improved sanitation conditions or behavioral adaptation by residents. However, this study cannot establish direct causality because complaint volume alone does not measure municipal response effectiveness. Instead, the findings demonstrate that public complaint systems reflect complex interactions between environmental conditions, visibility, and citizen behavior.

Limitations

Several limitations should be acknowledged. First, the analysis focuses on complaint volume rather than complaint resolution or municipal response time. Second, the models do not include additional control variables such as borough, weather severity, or population density. Third, daily observations may exhibit temporal autocorrelation and finally, overdispersion suggests additional unexplained variability within the dataset.

Conclusion

This study used Poisson and quasi-Poisson regression models to examine how major snowstorms affect dog waste complaints in New York City. The results demonstrate that snowfall significantly alters complaint behavior, though the direction and magnitude of the effect vary substantially across storm events. The January 2022 and January 2026 storms both produced increases in post-storm complaints, supporting the theory that snowfall temporarily conceals waste before exposure increases during melting periods. The January 2026 storm produced the strongest effect, with complaint counts increasing by approximately 289%. In contrast, the February 2026 storm produced a 55% decrease in complaints after snowfall, indicating that complaint patterns are not uniform across events. Overall, the study demonstrates the usefulness of NYC 311 complaint systems for analyzing behavioral and environmental changes during periods of urban disruption.

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